

CSS



```
body {  
  font: x-small  
  background: #  
  color: black;  
  margin: 0;  
  padding: 0;
```

Outline

1. CSS Syntax
2. CSS Selectors
3. Basic Text Styles
4. Margin, Border and Padding

CSS Syntax

↓ Selector

```
.my-css-rule {  
  background: red;  
  color: beige;  
  font-size: 1.2rem;  
}
```

Declaration →

Property Value

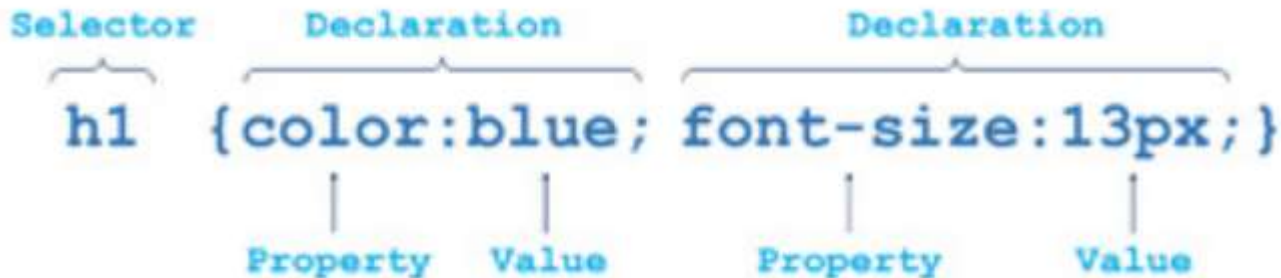
A diagram illustrating the components of a CSS rule. The rule is: `.my-css-rule { background: red; color: beige; font-size: 1.2rem; }`. The selector `.my-css-rule` is highlighted in a light blue box, with a blue arrow labeled "Selector" pointing to it. The entire rule body is enclosed in a light grey box. Inside, the declaration `color: beige;` is highlighted in a light green box, with a blue arrow labeled "Declaration" pointing to it. Within this declaration, `color:` is in a light red box and `beige;` is in a light blue box. A blue arrow labeled "Property" points to `color:` and a blue arrow labeled "Value" points to `beige;`. The other declarations, `background: red;` and `font-size: 1.2rem;`, are not highlighted.

CSS – Cascading Style Sheets

- CSS is used to control the **presentation** (*look and feel*) and **layout** (*positioning*) of web page elements
- Allows **separating** web page content from its design and **visual appearance**
- Used in conjunction with HTML
 - **HTML** is used for describing the **content** of a web page, **CSS** is used for describing its **presentation**
- CSS a flexible, cross-platform, **standards-based** styling language developed by the **W3C**

Style Sheets Syntax

- Stylesheets consist of **rules** that describe the styling to be applied
 - Each rule has **selectors** and **declarations**. A declaration specifies a **property** and its **value**



- Declarations are separated by semicolons
- Properties and values are separated by colons
- Comment in CSS `/* comment */`

Ways to incorporate CSS in an HTML document

- **Inline** – style included as the attribute of an HTML tag:

```
<p style="color:sienna;margin-left:20px;">This is a paragraph.</p>
```

- **Embedded** – CSS rules are contained in the head section:

```
<head>
<style>
p {color:sienna; margin-left:20px;}
body {text-align:center;}
</style>
</head>
```

- **External** - separate .css file referenced in the HTML:

HTML source code:

```
<head>
<link rel="stylesheet" href="main.css">
</head>
```

'main.css':

```
p {color:sienna;
margin-left:20px;}
body {text-align:center;}
```

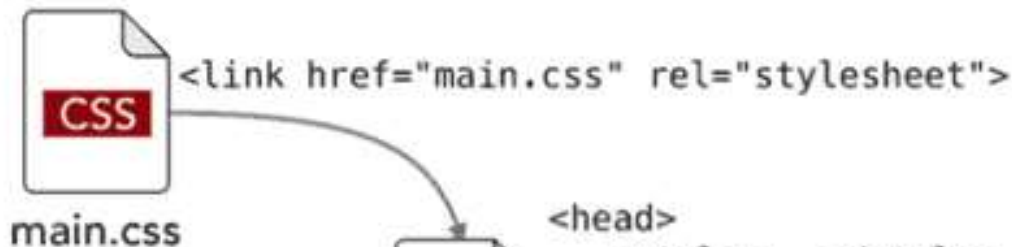


- ✓ Ensure
- ✓ Improve **consistent look and feel**, **reusability** and **maintainability**

The “Cascade”

- The **last rule applied wins**: Styles are applied in the order that they're found, and last styles always overwrite earlier styles in case of a conflict

```
<link rel="stylesheet" href="styles/qu-styles.css">  
<link rel="stylesheet" href="styles/ceng-styles.css">  
<link rel="stylesheet" href="styles/cse-styles.css">
```



cse-styles will win in case of a conflict

inline styles will win in case of a conflict

`<p style="font-size: .9em; margin-bottom: 1.2em;">`

index.htm

Inheritance

- Inheritance means that child elements will **inherit** the properties applied to a parent element


main.css

```
body {font-family: Arial;  
      font-size: 100%;  
      color: gray;}  
  
h1 {color: #900}
```


index.htm

```
<html>  
  <head>...</head>  
  <body>  
    <h1>Main heading</h1>  
    <p>This is body copy.</p>  
    <h2>This is a subheading</h2>  
    <p>More body copy</p>  
  </body>  
</html>
```

Main heading

This is body copy.

This is a subheading

More body copy

- Body selector has a font, font-size, and color applied to it:
 - Every element **inherits** those styles from the body selector
 - The style applied to h1 **overrides** the inherited color => h1 element appears in red berry color

CSS Selectors

Selectors: used to select elements to style on an HTML page

- **Element Selectors** - matches a HTML element directly

- Apply page-wide

- e.g., `p { font-family: verdana; }` applies the style to all `<p>` tags

- **Class Selectors**



- Defines a **named** style (prefix the name with dot (.))

- Can apply to any page element using the class attribute

- e.g., `.redBorder {border: 1px solid red}` defines a style named redBorder

- `<p class='redBorder'>Using the class attribute to apply the redBoder style to this paragrph</p>`

- **ID Selectors**



- Apply to one specific tag

- Use hash (#) followed by the tag id to select the element to be styled

- Good for linking to specific part of a page

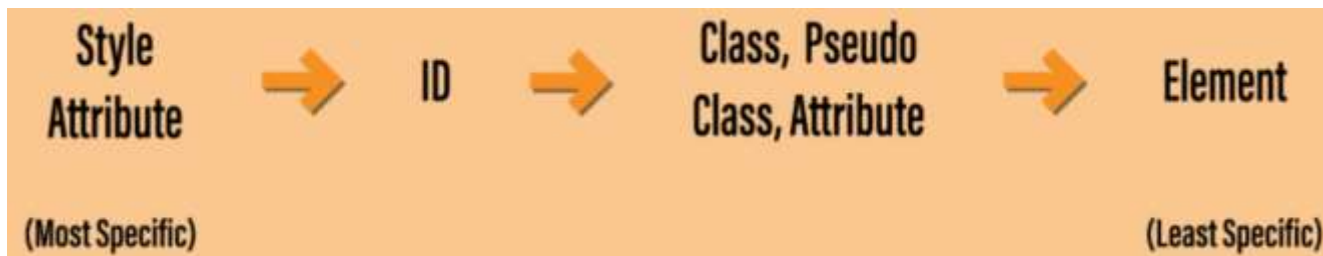
- e.g., `#errorMsg { color: red; }` apply the style to the element with id `errorMsg`

Specificity

```
<h1 id="my-heading" class="red">What colour am I?</h1>
```

```
.red {  
  color: red;  
}  
  
#my-heading {  
  color: blue;  
}  
  
h1 {  
  color: green;  
}
```

- We have 3 selectors for the h1 with different colors
 - What will the color be?
=> h1 would be **blue**
because the ID selector is more specific than the others
- The style for a **more specific selector** will be **applied** to the element



Universal

- [*]: Selects all elements
 - Used to override the browser default styles

```
* {  
    font-family: "Times New Roman";  
    color: blue;  
}
```

An element can have more than one class

```
<h1 class="red bold">Red and Bold</h1>
<h1 class="blue bold">Blue and Bold</h1>
<h1 class="bold">Bold</h1>
<h1 class="blue">Blue</h1>
```

```
.red {
  color: red;
}

.blue {
  color: blue;
}

.bold {
  font-weight: bold;
}
```

- Class Selector selects all elements whose **class attribute** matches the selector
 - The selector starts with a period (.) followed by the class name
 - An element can have more than one class

Combined Selectors

element, element	div, p	Selects all <div> elements and all <p> elements
element element	div p	Selects all <p> descendant elements inside <div> elements
element > element	div > p	Selects all <p> elements that are direct child of <div> element

e.g.,

li a {text-decoration: none}

- This will match all <a> tags that are inside of

See more combinations here:

https://www.w3schools.com/cssref/css_ref_combinators.php

https://www.w3schools.com/cssref/css_selectors.asp

Attribute Selectors

- Selects the elements whose attribute-value pair matches the selector

[attribute ^ = value]	a[href ^ = "https"]	Selects every <a> element whose href attribute value begins with "https"
[attribute \$ = value]	a[href \$ = ".pdf"]	Selects every <a> element whose href attribute value ends with ".pdf"
[attribute * = value]	a[href * = "qu"]	Selects every <a> element whose href attribute value contains "qu"

https://www.w3schools.com/cssref/css_selectors.asp

Pseudo-classes

- **Pseudo-classes** are used to define an **element state**. E.g.,
 - Style an element when a mouse is over it (`:hover`)
 - Style visited and unvisited links differently (`:visited` , `:link`)
 - Style an element when it gets focus (`:focus`)

e.g.,

`a:hover { color: red; }` -> Style link on mouse over

```
/* unvisited link */
a:link {
  color:  firebrick;
}

/* visited link */
a:visited {
  color:  limegreen;
}
```

```
input:focus {
  background-color:  yellow;
}
```


Structural Pseudo-classes

:first-child	tr:first-child	First row of an HTML table
:last-child	tr:last-child	Last row of an HTML table
:nth-child(n)	tr:nth-child(2)	Second row of an HTML table
:nth-last-child(n)	tr:nth-last-child(2)	Second row of an HTML table, counting from the last row
:nth-child(odd)	tr:nth-child(odd)	Every odd row of an HTML table
:nth-child(even)	tr:nth-child(even)	Every even row of an HTML table

<https://www.w3schools.com/cssref/trysel.asp>

has selector

- CSS **:has()** selector allows us to style an element based on its descendants
- e.g., assign a green background to any div having p as its descendent

```
div:has(p) {  
  background-color:  limegreen;  
}
```

Pseudo-elements

- A CSS pseudo-element is used to style specified parts of an element. E.g.,
 - Style the first letter or line of an element

```
p::first-letter { color:  #ff0000; }  
p::first-line { color:  #0000ff; }
```

- **Insert content** around the selected element

```
p::before { content: "«"; }
```

Insert « before the content of each <p> element

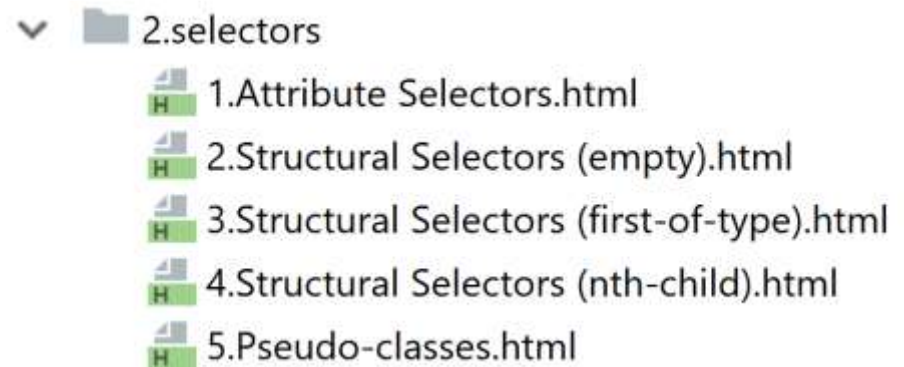
```
p::after { content: "»"; }
```

Insert » after the content of each <p> element

Selectors Summary

- A style consists of a selector, followed by property/value pairs
- Selectors:
 - Element Selectors
 - Class Selectors
 - ID Selectors
 - Combined Selectors
 - Attribute selectors
 - Pseudo-classes
 - Structural pseudo-classes
 - Pseudo-elements

Examples



Basic Text Styles

Text-related CSS Properties

- **color** – specifies the color of the text
- **font-size** – size of font: `xx-small`, `x-small`, `small`, `medium`, `large`, `x-large`, `xx-large`... or numeric value
- **font-family** – comma separated font names
 - Example: `verdana`, `sans-serif`, ...
 - The browser loads the first one that is available
- **font-weight** can be `normal`, `bold`....

Text-related CSS Properties (2)

- **font-style** – styles the font
 - Values: **normal, italic, oblique**
- **text-decoration** – decorates the text
 - Values: **none, underline, line-through...**
- **text-align** – defines the alignment
 - Values: **left, right, center, justify**

Font Sizing

Two ways to define font sizes in CSS

- **Fixed Font Sizing**: defines the size of fonts using absolute units such as points (**pt**) or pixels (**px**)
- **Relative Font Sizing**: defines the size of fonts using relative units such as **em** and **rem** units
 - This allows fonts to scale appropriately to different resolutions, browsers or platforms
 - **em** is relative to the font-size of its direct or nearest parent
 - **rem** is relative to the font-size of the html (root) element

Relative Font Sizing

```
body {font-size: .8em;  
      font-family: Verdana, Helvetica, Sans-Serif;}  
h1    {font-size: 1.2em;}  
div   {font-size: .8em;}  
li     {font-size: .8em;}  
p     {font-size: .8em;}
```

03.text-styles

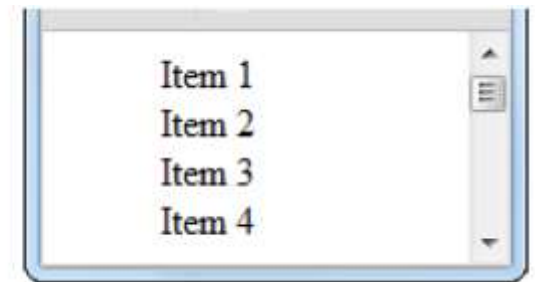
 1.font-size.html

 2.em-vs-rem.html

Styles for Lists

- List properties are used to define the look and feel of the list items
 - Values for `<ul>`: `circle`, `square`,...
 - Values for `<ol>`: `upper-roman`, `lower-alpha`
 - Values for both: `none`

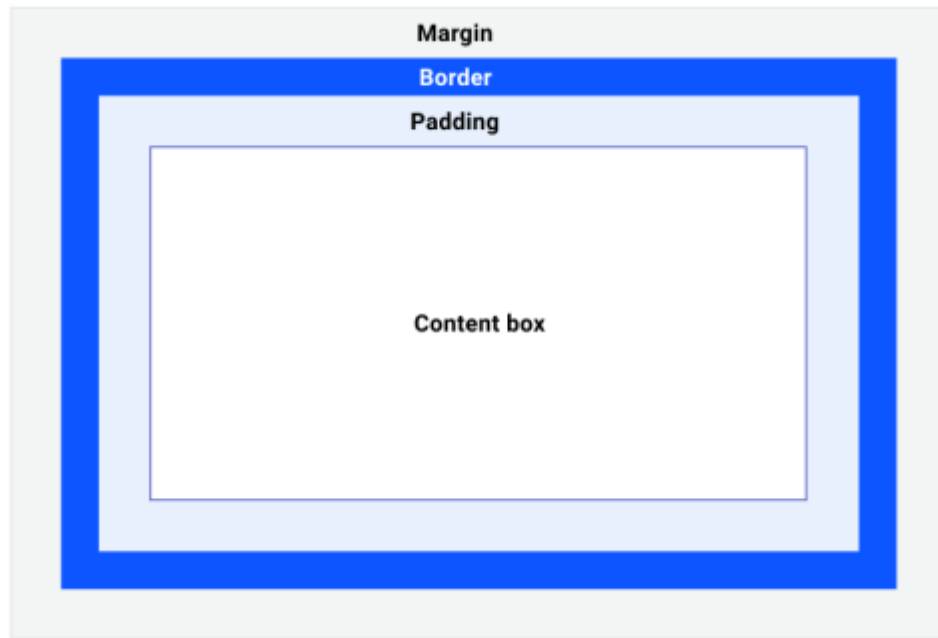
```
ul  
{  
    list-style-type:none;  
}
```



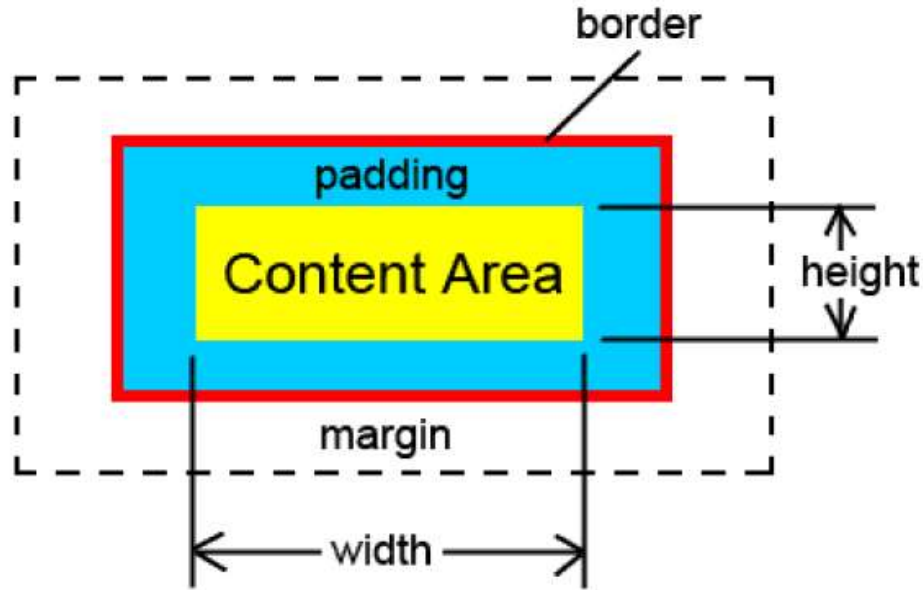
Practice ...

- Use the W3Schools try-it-yourself editor to try styling each of these properties
- **Background**
http://www.w3schools.com/css/css_background.asp
- **Text**
http://www.w3schools.com/css/css_text.asp
- **Fonts**
http://www.w3schools.com/css/css_font.asp
- **Lists**
https://www.w3schools.com/css/css_list.asp

Margin, Border and Padding



Box Model



- Each tag is a box, and its properties can be styled:
 - **Margin** - the space that separates the boxes
 - **Border** - the line around each edge of the box
 - **Padding** - the space between the border and the contents

Margin and Padding

- Margin and padding define the spacing around the element
 - Numerical value, e.g. **0.8rem**
 - Can be defined for each of the four sides separately: **margin-top**, **padding-left**, ... or using short rules:
- **margin: 0.8rem;**
 - Sets all four sides to have margin of **0.8rem**
- **margin: 10px 20px;**
 - top and bottom to 10px, left and right to 20px
- **margin: 1px 3px 5px 7px;**
 - top, right, bottom, left (clockwise from top)
- Same for padding

Borders

- Border style:

`border-width: 1px;`

`border-style: solid;`

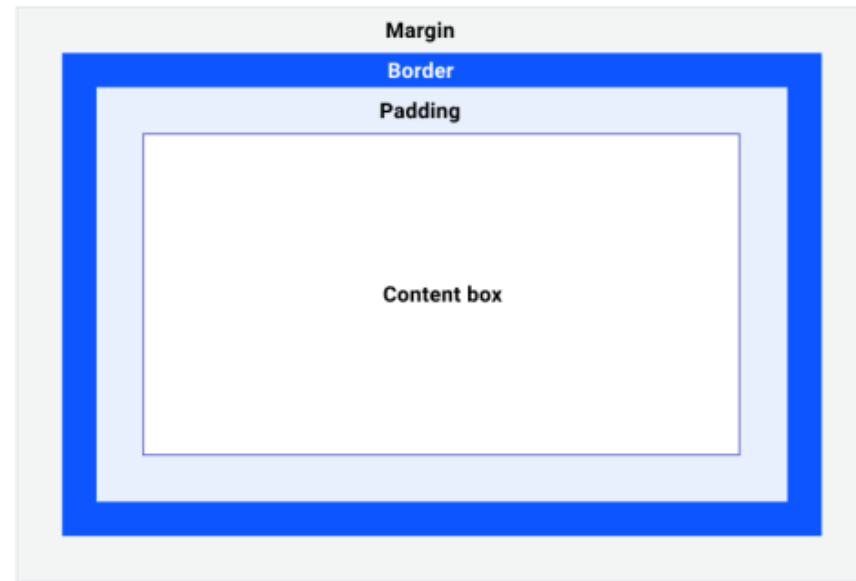
`border-color: red;`

- **border-width:** `thin`, `medium`, `thick` or numerical value
- **border-color:** `color alias` or `RGB value`
- **border-style:** `none`, `dotted`, `dashed`, `solid`, `double`, ...

- Shorthand rule for setting border properties:

`border: 1px solid red;`

- Can specify different borders for the sides using:
`border-top`, `border-left`, `border-right`, `border-bottom`



Rounded Corner & Centered Box



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```
.centered-rounded-box {  
  display: flex;  
  column-gap: 1rem;  
  align-items: center;  
  
  padding: 0.5rem;  
  border: 1px solid saddlebrown;  
  border-radius: 15px;  
  
  width: 80%;  
  margin: auto;  
}
```

Rounded Corner
box

Centered box

References

- CSS Tutorials <http://www.w3schools.com/css/>
- CSS Course <https://web.dev/learn/css/>
- Cheat sheet <https://htmlcheatsheet.com/css/>
- CSS developer guide
<https://developer.mozilla.org/en-US/docs/Web/Guide/CSS>
- Selectors
<http://code.tutsplus.com/tutorials/the-30-css-selectors-you-must-memorize--net-16048>